

A House Divided: Modern Heterogeneous-Effects Estimators and the Contested Mortality Effects of Hospital Desegregation, 1959–1988

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Abstract

Almond, Chay, and Greenstone (2006) argued that the Title VI Medicare-certification campaign of 1966–1968 produced a large decline in Black postneonatal mortality in the rural South. Anderson, Charles, and Rees (2024) revisited the question with a county-of-nearest-hospital design and reported a statistically null effect. We apply five staggered difference-in-differences estimators — two-way fixed effects (TWFE), Callaway–Sant’Anna (CS), Borusyak–Jaravel–Spiess (BJS), de Chaisemartin–D’Haultfoeulle (dCDH), and Sun–Abraham (SA) — to a unified eleven-state Confederate-South panel of Black postneonatal mortality, using the `cert_75` treatment definition of Anderson, Charles, and Rees (2024). We additionally construct a federal-source-uniform extended panel back to 1959 by combining NCHS Mortality Detail File microdata with audited VSUS Vol I Natality tabulations; the extended panel is the primary specification on which we anchor inference.

On the federal-source-uniform extended panel (1959–1988), the four heterogeneity-robust estimators all reject zero at the 5-percent level: TWFE -1.85 $[-3.43, -0.28]$, CS -5.09 $[-7.05, -3.14]$, BJS -2.41 $[-3.87, -0.95]$, dCDH -4.07 $[-6.12, -2.02]$. Sun–Abraham returns $+0.84$ $[-0.33, +2.00]$, an apparent sign flip whose interval contains zero and which we trace to the dominant 1967 cohort’s contribution to SA’s cohort-share aggregator. A selective-fertility bound from Thompson (2024) Table 4 implies a mechanical compositional shift of approximately -2.7 per 1,000; the CS and dCDH estimates exceed this bound and are consistent with a Title-VI-causal effect not exhausted by fertility selection, while the TWFE and BJS estimates fall just below it and are not separately distinguishable from a pure compositional mechanism. Because the Black fertility decline of 1965–1968 is itself partly endogenous to civil-rights enforcement, the bound separates *non-fertility-mediated* Title-VI channels from the rest rather than direct treatment effects from no treatment effect.

On the public-CMF window (1968–1988), reported as a robustness comparison, the same four heterogeneity-robust estimators report negative ATTs between -1.3 and -2.5 per 1,000 live births: TWFE -1.30 $[-3.00, +0.40]$, CS -2.50 $[-4.45, -0.55]$, BJS -1.85 $[-3.87, +0.17]$, dCDH -2.27 $[-4.18, -0.36]$. CS and dCDH reject zero on the shorter panel; TWFE and BJS do not. The attenuation toward zero on the shorter panel is consistent with two diagnostic findings: (i) a Goodman-Bacon decomposition attributes 36 percent of TWFE weight to “forbidden” later-vs-already-treated comparisons — comparisons in which a unit treated later is benchmarked against a unit treated earlier acting as the implicit control, which delivers biased treatment-effect estimates under treatment-effect heterogeneity (Goodman-Bacon, 2021) — and the mean of those comparisons (-0.52) is much less negative than the clean treated-vs-never-treated comparisons’ mean (-2.44); and (ii) the dominant 1967 cohort has zero pre-period observations within the public NBER CMF, so

the staggered-DiD estimators must extrapolate the counterfactual from late-cohort variation that coincides with substantially different state-level economic and policy environments. The extended panel removes the cohort-anchoring problem; the heterogeneity-robust estimators address the forbidden-comparison problem; together they produce the larger magnitudes that the heterogeneity-robust estimators converge on.

The post-2018 heterogeneity-robust estimator menu produces a *frontier* of estimates rather than a single point. On the federal-source-uniform extended panel and after the selective-fertility adjustment, the data favors a substantively negative effect of Title VI hospital desegregation on Black postneonatal mortality consistent with the original claim of Almond, Chay, and Greenstone (2006), with magnitudes between -1.9 and -5.1 per 1,000 depending on estimator. The Anderson, Charles, and Rees (2024) null is recoverable on the public-CMF window for the TWFE and BJS specifications, both of which the cohort-anchoring diagnostics flag as attenuated, suggesting that the original disagreement reflects in substantial part the limits of post-1968-only public mortality data.

Keywords: Title VI, hospital desegregation, Black infant mortality, staggered difference-in-differences, Callaway–Sant’Anna, Goodman–Bacon decomposition, Honest-DiD.

1. Introduction

The 1964 Civil Rights Act was operationally enforced in southern hospitals through Medicare certification. Beginning July 1, 1966, the Department of Health, Education, and Welfare’s Office of Equal Health Opportunity (HEW–OEHO) required, as a condition of receiving federal Medicare reimbursement, that hospitals certify nondiscriminatory treatment of Black patients and staff under Title VI [reynolds1997federal; smith1999healthcaredivided]. The campaign was, in Reynolds’s (1997) characterization, “one of the most successful civil-rights enforcement operations of the federal government.” Within eighteen months, 96 percent of the Deep South’s hospitals were certified; the remainder either closed or accepted formal segregation and forfeited federal funds.

Twenty years later, Almond, Chay, and Greenstone (2006) [hereafter ACG] used the NBER Compressed Mortality File and the staggered 1966–1968 county-level certification rollout to argue that this campaign produced a large, persistent decline in Black postneonatal mortality across the rural South. The ACG paper, circulated as MIT Working Paper 07-04 in 2007 but never published in a peer-reviewed journal, became the canonical empirical reference for the proposition that civil-rights enforcement saved tens of thousands of Black infant lives in the late 1960s. It is cited in virtually every survey of post-1960s mortality declines and has been influential in establishing the broader narrative that Black–White health gaps in the United States closed substantively only when federal civil-rights enforcement gave Black Americans physical access to white-only health-care infrastructure.

Anderson, Charles, and Rees [hereafter ACR] revisited the question in a 2024 paper now forthcoming at the *Review of Economics and Statistics*. ACR (a) compiled the underlying hospital-certification dates directly from the annual *Journal of the American Hospital Association* “Guide Issue” rather than relying on the secondary sources that ACG had used; (b) coded county-level “access” using four different bed-coverage thresholds (any certification, at least 25 percent, at least 50 percent, at least 75 percent of pre-1965 hospital beds); and (c) restricted attention to the five Deep-South states most exposed to the certification campaign. Their headline conclusion, on the at least 75-percent bed-coverage binding (`cert_75`), is that the campaign had no statistically detectable effect on Black postneonatal mortality. ACR (2024) is the first careful re-examination of ACG’s headline claim and reports a *null* on essentially the same outcome.

The disagreement matters for two reasons. First, the foundational claim that civil-rights enforcement produced a large, measurable mortality dividend is uncomfortable to lose; the absence of a peer-reviewed Almond–Chay–Greenstone paper has meant that the field has cited the working-paper version for two decades as if it were settled. Second, the methodological texture of the disagreement is informative for a broader question: how should the discipline now read the large 2000s and 2010s difference-in-differences literature that relied on the staggered-TWFE estimator? Goodman–Bacon (2021), Callaway and Sant’Anna (2021), Sun and Abraham (2021), de Chaisemartin and D’Haultfœuille (2020), Borusyak, Jaravel, and Spiess (2024),

and Rambachan and Roth (2023) have collectively documented that staggered-TWFE estimators can deliver biased estimates of average treatment effects when treatment timing varies, treatment effects are heterogeneous, or pre-trends are present. None of those papers has been applied side-by-side to the ACG–ACR question.

This paper does so. We construct a unified eleven-state Confederate-South county-year panel for the period 1968–1988 — the public-NBER-CMF window — and apply five staggered-DiD estimators (TWFE, Callaway–Sant’Anna, Borusyak–Jaravel–Spiess, de Chaisemartin–D’Haultfoeulle, and Sun–Abraham) to the same outcome (Black postneonatal mortality per 1,000 live births), the same treatment definition (the ACR `cert_75` binding), and the same sample frame. We then extend the panel backward to 1959 by transcribing pre-1968 state vital-statistics tabulations — the same data source that ACG (2006) used for their pre-period and that ACR (2024) acknowledged as essential for the dynamic event-study but did not themselves transcribe. The extended-panel estimates reveal whether the original disagreement was driven by the post-1968-only NBER CMF window or by something else.

Our contribution is methodological adjudication, not a new empirical claim about whether hospital desegregation reduced Black infant mortality. We document the *shape* of the estimator frontier on this canonical question — under what data window, under what estimator, the data favors the ACG reading; under what conditions the data favors ACR; and what the partial-identification frontier looks like under Rambachan–Roth bounds. We hope the exercise is useful both as a substantive adjudication of one of the most important questions in modern U.S. health-equity history and as a methodological case study for how to read older staggered-DiD literatures in light of the post-2018 econometric developments.

The paper proceeds as follows. Section 2 reviews the historical and econometric literature. Section 3 documents the data. Section 4 describes the five estimators and the robustness suite. Section 5 reports results. Section 6 discusses the partial-identification frontier and what would be needed to pin the magnitude. Section 7 concludes.

2. Background

2.1 The Title VI hospital-desegregation campaign — institutional and political history

The 1964 Civil Rights Act outlawed discrimination by recipients of federal funds, but its operative enforcement in southern hospitals required a separate instrument: the federal money attached to Medicare. Before 1966, southern hospitals received federal Hill-Burton construction grants without operative Title VI compliance. The 1946 Hill-Burton Hospital Survey and Construction Act explicitly authorized “separate but equal” facilities — Section 622(f) of the Act permitted federal funds to flow to hospitals that maintained racial separation provided the separate Black facilities were “of like quality.” Between 1946 and 1965, the federal Hill-Burton program funded approximately 100 segregated Black hospitals across the South and provided construction grants to hundreds of additional white-only facilities [smith1999healthcaredivided; reynolds1997federal].

The constitutional challenge to Hill-Burton’s separate-but-equal clause culminated in the 1963 Fourth Circuit decision in *Simkins v. Moses H. Cone Memorial Hospital* (323 F.2d 959, 4th Cir. 1963), in which the court ruled that the Hill-Burton separate-but-equal provision was unconstitutional under the Equal Protection Clause and that hospitals receiving federal funds could not maintain racial segregation. The decision applied to the Fourth Circuit states (NC, SC, VA, WV, MD) but not to the Fifth Circuit states (AL, FL, GA, LA, MS, TX). The Supreme Court denied certiorari in 1964 (376 U.S. 938), leaving *Simkins* as Fourth-Circuit-only precedent until the broader 1964 Civil Rights Act and the 1966 Medicare Act enforcement.

The operative enforcement instrument was Title VI of the 1964 Civil Rights Act applied through Medicare certification, beginning July 1, 1966. Under the Medicare Act of 1965 (Public Law 89-97), hospitals receiving Medicare reimbursement were required to certify nondiscrimination in admissions, room assignments, staff privileges, and physician practice. The certification process was operationally managed by the Department of Health, Education, and Welfare’s Office of Equal Health Opportunity (HEW–OEHO), established within the Public Health Service. HEW–OEHO compliance officers conducted on-site inspections of southern hospitals

during February 1966 through October 1968, verifying that hospitals had eliminated racially-designated wards, integrated staff dining facilities, opened medical-staff appointments to Black physicians, and complied with the Title VI nondiscrimination requirements.

The political economy of the certification process was complex. Hospital trustees in the Deep South initially resisted certification — 214 hospitals in the five-state Deep South refused federal Medicare reimbursement rather than integrate. The economic pressure of forgoing Medicare reimbursement, however, proved decisive: within twelve months of the July 1, 1966 effective date, approximately 96 percent of Deep-South hospitals had certified. The laggards were predominantly small, rural facilities with limited Medicare-eligible patient bases and limited financial reserves; they either closed (~30 closures in 1966–1969) or reluctantly certified. The political-historical literature [reynolds1997federal; smith1999healthcaredivided; smith2005race; largent2018segregation] documents that the certification process was conducted under substantial federal-state political tension, with state governors and southern members of Congress repeatedly attempting to block or delay HEW–OEHO compliance enforcement.

The HEW–OEHO compliance files themselves are archived at the National Archives and Records Administration as NARA Record Group 235, Series 5 (Health Equity Compliance Files, 1966–1972). Reynolds (1997) and Largent (2018) have separately conducted archival work in these files; their accounts of the operational compliance process — site inspections, formal compliance reviews, hospital-by-hospital certification negotiations — are the primary source basis for our understanding of cosmetic-compliance heterogeneity. Smith (1999, 2005) documents that many certified hospitals dual-listed wards as “integrated” while continuing de facto segregation through admission patterns, room-assignment defaults, and physician-practice restrictions. The `cert_75` binary indicator in our analytic panel averages over this substantial residual segregation.

The campaign’s operational success was measured at the federal level through the “Guide Issues” of the *Journal of the American Hospital Association*, published annually in February and October. The Guide Issue listed, for each hospital, whether it had been “certified for participation in the Health Insurance for the Aged (Medicare) Program by the Department of Health, Education, and Welfare” as of the publication’s reference date. Anderson, Charles, and Rees (2024) hand-coded the Guide Issues for the five Deep-South states (AL, GA, LA, MS, SC) for 1967–1974 to reconstruct hospital-level certification dates, then aggregated to the county-of-nearest-hospital level under four alternative bed-coverage thresholds: any certification (`cert_any`), at least 25 percent of pre-1965 county hospital beds (`cert_25`), at least 50 percent (`cert_50`), and at least 75 percent (`cert_75`). The resulting county-year certification panel — reported in their Appendix Table B1 — is the analytic foundation for the modern reassessment of the ACG findings.

The dispersion in `cert_75` is meaningfully wider than in `cert_any`. In our merged panel, 350 of 415 Deep-South counties with assigned dates were certified in 1967 under `cert_75`; 43 in 1968; 17 in 1969–1971; and 5 in 1972 or later. Under `cert_any` (any hospital certified), the distribution is much more concentrated in 1966–1967. The dispersion in `cert_75` is exactly the variation that ACR (2024) argue powers the modern county-of-nearest-hospital identification strategy and that ACG (2006) treated as too modest to exploit.

2.2 The Almond–Chay–Greenstone (2006) headline

ACG used the NBER Compressed Mortality File 1959–1988 (which extends backward via state vital-statistics tabulations to 1959) and a staggered-DiD design at the county level. Their preferred specification compared Black postneonatal mortality rates in southern counties before and after Medicare certification of the county’s hospitals, controlling for county and year fixed effects and Black–White rates. They reported a ~5–6 per-1,000-births reduction in Black postneonatal mortality, concentrated in the rural Deep South and in cohort-1967 counties. The ACG estimate implies on the order of ~25,000 Black infant lives saved over the 1965–1972 period from a single federal policy intervention.

ACG (2006) was never peer-reviewed. The paper has nevertheless been cited several thousand times and is the standard reference for the proposition that Title VI enforcement produced a large mortality dividend. It is the empirical anchor for the broader narrative that the Civil Rights Act materially extended Black life expectancy in the late 1960s.

2.3 The Anderson–Charles–Rees (2024) null

ACR reproduced ACG’s panel structure with three differences. First, they hand-coded hospital certification from primary sources (the JAHA Guide Issues) rather than relying on the secondary derivative dates ACG used. Second, they restricted attention to the five Deep-South states most exposed to the campaign rather than the eleven-state Confederate sample ACG had used. Third, they reported four alternative bed-coverage thresholds rather than ACG’s single any-certification binary. Across all four thresholds and on essentially the same outcome (Black postneonatal mortality), ACR reported a statistically null effect.

ACR’s interpretation is that the ACG result was an artifact of the secondary certification-date measure ACG used, which conflated county-level access with state-level certification timing. The implicit methodological diagnosis is that ACG had a high false-positive rate driven by measurement error in the treatment.

2.4 Adjacent literatures on civil-rights and Black mortality declines

The civil-rights-era reduction in Black infant mortality is one of the most-studied phenomena in twentieth-century American population health. Beyond the direct hospital-desegregation channel that ACG and ACR debate, four adjacent literatures bear on the interpretation of the headline ATT.

First, the federal War on Poverty health programs of 1965–1972 — Medicaid (1965), Community Health Centers (Bailey and Goodman-Bacon 2015), and Maternal and Child Health Title V expansions — produced contemporaneous mortality reductions that overlap the Title VI exposure window. Goodman-Bacon (2018 *JPE*) identifies a 1965 Medicaid implementation effect on elderly mortality; Bailey and Goodman-Bacon (2015 *AER*) identify CHC effects on older-adult mortality 1965–1974; Currie and Gruber (1996 *JPE*) and (1996 *QJE*) identify Medicaid expansions and infant mortality more broadly. The methodological challenge in this paper is to separate the Title-VI-specific hospital-integration channel from the broader War-on-Poverty health-financing channel that operated through the same calendar window.

Second, the voting-rights and political-economy literature — Cascio and Washington (2014 *QJE*) on the Voting Rights Act and state funds — documents that Black political enfranchisement after 1965 produced its own measurable effects on state-level resource allocation to majority-Black areas, which would have its own infant-mortality consequence. Our covariate-adjusted robustness (§5.8) adds VRA Section 5 preclearance as a state-year constant; the substantive effect on the headline ATT is small (at most 0.06 per 1,000), but the VRA channel is methodologically a confounding policy.

Third, the selective-fertility literature (Thompson 2024 *JHR*, Aizer-Currie-Moretti-Yang 2014, Eli-Currie 2019) documents that the Black fertility decline of 1964–1970 was concentrated in high-mortality-risk pregnancies, producing a mechanical reduction in measured infant mortality that any of our five estimators would attribute to treatment. Our §6.3 quantitative bound on selective fertility addresses this confound directly; the bound is approximately -2.7 per 1,000 from compositional shift alone, comparable in magnitude to the TWFE and BJS estimates and smaller than the CS and dCDH estimates on the extended panel.

Fourth, the broader Black mortality literature documents secular and policy-driven declines in Black mortality across the lifespan during this period. Chay and Greenstone (2003 *QJE*) on air-quality regulation; Cutler and Miller (2005 *Demography*) on twentieth-century water-quality interventions; Bleakley (2007 *QJE*) on hookworm-eradication effects on cohort outcomes; Aizer, Eli, Ferrie, and Lleras-Muney (2016 *AER*) on Mothers’ Pensions effects on Black mortality; Almond and Mazumder (2011 *AEE:Applied*) on prenatal-care effects; Brown, Kowalski, and Lurie (2020 *RFS*) on Medicaid expansions and infant mortality; Currie and Schwandt (2016 *Science*) on the post-1965 Black–White infant-mortality convergence. Each of these literatures intersects the Title VI exposure window through different mechanisms; the multi-confound identification problem is substantial.

2.5 The staggered-DiD econometric literature

The methodological foundation for this paper is the post-2018 staggered-DiD literature that documents the limits of the two-way-fixed-effects estimator under treatment-effect heterogeneity. The foundational result is Goodman-Bacon (2021 *Journal of Econometrics*), which decomposes the TWFE estimator into

2×2 difference-in-differences with non-negative-and-non-negative-weights depending on the comparison type, and demonstrates that “forbidden” later-vs-already-treated comparisons can carry negative weight when treatment effects vary across cohorts or over event time.

The post-Goodman-Bacon literature proposes alternative estimators that avoid the forbidden-comparison problem. Callaway and Sant’Anna (2021 *Journal of Econometrics*) propose an ATT(g,t) framework using clean comparisons (treated vs. never-treated or treated vs. not-yet-treated) aggregated with non-negative weights. Sun and Abraham (2021 *Journal of Econometrics*) propose a cohort-saturated event-study interacting cohort-fixed-effects with relative-time dummies. de Chaisemartin and D’Haultfœuille (2020 *AER*; 2024 *RES*) propose switching-on event-studies comparing treated and stably-untreated units. Borusyak, Jaravel, and Spiess (2024 *Review of Economic Studies*) propose an imputation estimator that fits TWFE on untreated observations only. Athey and Imbens (2022 *Journal of Econometrics*) develop design-based identification under random assignment of treatment timing.

The partial-identification frontier under violations of strict parallel trends is documented in Rambachan and Roth (2023 *Review of Economic Studies*), which provides bounds for the post-treatment average under restrictions on the relative magnitude or smoothness of differential trends. The Honest-DiD framework has become standard for sensitivity reporting in modern staggered-DiD work. Roth, Sant’Anna, Bilinski, and Poe (2023 *Journal of Econometrics*) provide a comprehensive review.

The methodological texture of these estimators is the principal contribution of this paper. None of them has been applied side-by-side to the ACG–ACR adjudication in the published literature. The closest precedent is Tomar, Yu, and Spiess (2024 *Working Paper*), which applies a smaller estimator menu (CS + SA only) to a different civil-rights question (school desegregation); we do not duplicate their work but build on the methodological framing.

3. Data

We combine three publicly available data inputs to construct a county-year panel of Black and White postneonatal mortality covering the eleven Confederate-South states from 1959 through 1988.

Mortality and population. Death counts come from the National Center for Health Statistics Compressed Mortality File (CMF) 1968–1988, distributed by the National Bureau of Economic Research at <https://data.nber.org/cmfile/>. The CMF is the public county-level mortality aggregate used by both Almond, Chay, and Greenstone (2006) and Anderson, Charles, and Rees (2024) and is the analytic foundation for the U.S. mortality–civil-rights literature [atgoodmanbacon2018publicinsurance; @bailley2015waronpoverty; @chay2003impact]. Each record in the mortality file is a count of deaths in a state × county × year × race-sex × age × ICD-cause cell. The file uses ICD-8 codes 1968–1978 and ICD-9 codes 1979–1988. Population denominators and live births in the same race-sex × age × county-year cells come from the companion NBER population file (pop6878 and pop7988, fixed-width format with logical record length 140 and twelve age-group population fields plus a live-births field).

Pre-1968 mortality and births. Because the NBER public CMF begins in 1968 and the pre-period cited by Almond, Chay, and Greenstone (2006) is therefore not in the CMF, we construct a federal-source-uniform pre-1968 extension. County-year Black and White infant, neonatal, and postneonatal death counts come from the NBER National Center for Health Statistics Mortality Detail File (MDF) for 1959–1967, the underlying federal microdata from which the published VSUS Vol II mortality tables are produced. We aggregate death records by state of residence, county of residence, race, and the NCHS detailed age code, mapping NCHS county codes to FIPS via the NBER 1990 crosswalk and an alphabetical-ordering decode for the 1959–1961 four-character coding scheme. Pre-1968 live births by county-year-race come from the VSUS Vol I Natality state tables; we audit county-year cells against state-year totals and set obviously implausible birth values (county births exceeding 50,000 or 50 percent of the state-year Black-births total) to missing. Construction details are in Appendix A.

Medicare-certification treatment dates. Hospital-level Medicare certification dates were originally compiled by Anderson, Charles, and Rees (2024) from the annual *Journal of the American Hospital Association*

“Guide Issues” 1967–1974. Anderson, Charles, and Rees (2024) aggregate these hospital-level certification dates to the county-of-nearest-hospital level under four bed-coverage thresholds and report the resulting 418-county Deep-South panel in their Appendix Table B1. We extract this Appendix Table B1 directly from the working-paper PDF and merge it onto the county-year panel by FIPS code. This gives treatment dates for all 418 Deep-South counties (Alabama, Georgia, Louisiana, Mississippi, South Carolina), of which 403 enter the analytic sample after dropping the 9 Alabama and 6 Georgia counties carrying Appendix B1 footnote exclusions.

Time-varying covariates. Medicaid implementation year [goodmanbacon2018publicinsurance Table 1] and Voting Rights Act Section 5 preclearance status [cascio2014valuingvote Table A1] enter as state-year constants. Hill-Burton funding [bailey2015waronpoverty], Community Health Center year [bailey2015waronpoverty], and AFDC monthly maximum [moffitt2003] are unavailable in machine-readable form at the county-year level and are documented as outstanding analyst-mediated access items.

Sample. Our analysis sample is the eleven Confederate-South states (Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, and Virginia). This is the Almond, Chay, and Greenstone (2006) sample frame; Anderson, Charles, and Rees (2024) restrict to the five Deep South subset (AL, GA, LA, MS, SC). We retain all eleven states so that we can decompose the Almond-Chay-Greenstone-Anderson-Charles-Rees disagreement into a *sample-frame* component (5-state vs. 11-state) and an *estimator* component (TWFE-style vs. heterogeneity-robust DiD). The public-CMF window panel contains 23,672 county-years (1,146 counties $\times$ 1968–1988); the federal-source-uniform extended panel additionally covers 9,723 pre-1968 cells (1959–1967).

Outcomes. Following ACG and ACR, our primary outcome is the Black postneonatal mortality rate, defined as $1000 \times (\text{Black } 28\text{--}365\text{-day deaths}) / (\text{live Black births})$ within a county-year cell. We also construct (i) Black neonatal mortality, (ii) Black infant mortality, (iii) the corresponding White rates and Black–White gaps, and (iv) Black motor-vehicle mortality (ICD codes 770/800) for cause-specific falsification.

Treatment. Treatment is staggered Medicare certification of the county’s hospital(s) under Title VI compliance, using ACR’s `cert_75` binding (first year by which Medicare-certified hospitals covered at least 75 percent of pre-1965 county hospital beds). In our merged panel, 350 of 415 Deep-South counties with assigned dates were certified in 1967; 43 in 1968; 17 in 1969–1971; and 5 in 1972 or later. The dispersion in `cert_75` is meaningfully wider than in the `cert_any` indicator.

4. Methods

4.1 Identification

We treat the staggered 1966–1968 rollout of Medicare-certification-driven Title VI compliance as a county-level natural experiment in hospital integration. The treatment date is the year by which Medicare-certified hospitals covered at least 75 percent of pre-1965 county hospital beds (`cert_75`). The outcome is the Black postneonatal mortality rate per 1,000 live births. The estimand of interest is the average treatment effect on the treated (ATT) on this outcome.

The contribution of this paper is to apply, on the *same* harmonised eleven-state Southern panel, the four leading heterogeneity-robust event-study estimators that have emerged in econometrics since 2018 — Callaway and Sant’Anna (CS, 2021), Sun and Abraham (SA, 2021), Borusyak, Jaravel, and Spiess (BJS, 2024), and de Chaisemartin and D’Haultfœuille (dCDH, 2020/2024) — and to compare them to the two-way-fixed-effects (TWFE) estimator used implicitly by Almond, Chay, and Greenstone (2006). We report all five estimators side-by-side as the headline of the paper.

4.2 Five estimators

The TWFE benchmark is

$$Y_{ct} = \alpha_c + \gamma_t + \beta \mathbf{1}\{t \geq G_c\} + \varepsilon_{ct},$$

where G_c is the `cert_75` year and α_c, γ_t are county and year fixed effects. The regression is weighted by Black live births and standard errors are clustered at the county level.

The Callaway–Sant’Anna estimator targets the group–time average treatment effect $ATT(g, t)$ and aggregates to a “simple” ATT and an event-time path under conditional parallel trends. We use the doubly-robust variant with never-treated controls. Implementation is via the Python `differences` package.

The Borusyak–Jaravel–Spiess estimator fits unit and time fixed effects on untreated observations only, then imputes the counterfactual $\hat{Y}_{ct}(\infty)$ for treated cells. Counties with no untreated observations (the 1967 cohort on the public-CMF window) are dropped from this step. Standard errors are cluster-bootstrapped at the county level ($B = 200$).

The de Chaisemartin–D’Haultfœuille estimator computes, for each calendar year t and horizon ℓ , the DiD between counties switching into treatment at t and stably-untreated counties. Standard errors are cluster-bootstrapped at the switching-event level ($B = 200$).

The Sun–Abraham estimator saturates the event-study with cohort-by-relative-time interactions and aggregates the cohort-specific event-time coefficients with cohort-size weights.

4.3 Diagnostics and robustness

Goodman-Bacon decomposition. We decompose the TWFE estimate into its constituent 2×2 DiD comparisons, reporting the share of TWFE weight attributable to (i) treated-vs-never-treated comparisons, (ii) earlier-treated-vs-later-treated comparisons, and (iii) the “forbidden” later-treated-vs-already-treated comparison.

Rambachan-Roth Honest-DiD. For each estimator’s post-treatment average over $\ell \in \{0, \dots, 5\}$, we compute partial-identification bounds under the relative-magnitude restriction $|\bar{\delta}_{\text{post}} - \bar{\delta}_{\text{pre,max}}| \leq \bar{M} \cdot \bar{\delta}_{\text{pre,max}}$ for $\bar{M} \in \{0, 0.25, 0.5, 1.0, 2.0\}$.

White-postneonatal placebo. Title VI desegregation should change Black hospital access without affecting White infants; any non-null effect indicates contamination from a non-Title-VI shock.

Cause-specific motor-vehicle placebo. We extract Black motor-vehicle deaths from the NBER CMF using IRECODE 770 (ICD-8) and 800 (ICD-9) and run all five estimators on Black motor-vehicle mortality per 100,000 population. Title VI desegregation has no plausible causal pathway to motor-vehicle mortality, so a null effect is the cleanest external falsification.

Leave-one-state-out. We re-estimate each estimator eleven times, dropping each Southern state in turn.

Covariate-adjusted robustness. We re-estimate each estimator with Medicaid implementation year and VRA Section 5 preclearance as time-varying covariates (Frisch–Waugh–Lovell residualization).

4.4 Implementation

All estimators are implemented in Python (`pyfixest` 0.40 for TWFE and SA; `differences` 0.2 for CS; manual implementations for BJS and dCDH). The full analysis reproduces end-to-end via `python3 analysis/run_all.py` in approximately ninety seconds.

5. Results

5.1 Headline static ATT — public-CMF window (1968–1988)

Figure 1 shows the treatment-timing distribution of the eleven-state Confederate-South sample under the `cert_75` binding: 304 counties enter in 1967, 48 in 1968, and the remainder spread between 1969 and 1974. Six of the eleven states (Arkansas, Florida, North Carolina, Tennessee, Texas, and Virginia) supply the never-treated control pool. The dominant 1967 cohort has no observed pre-period in the NBER Compressed

Mortality File, which begins in 1968 — the empirical constraint that motivates the federal-source-uniform extended panel in §5.7.

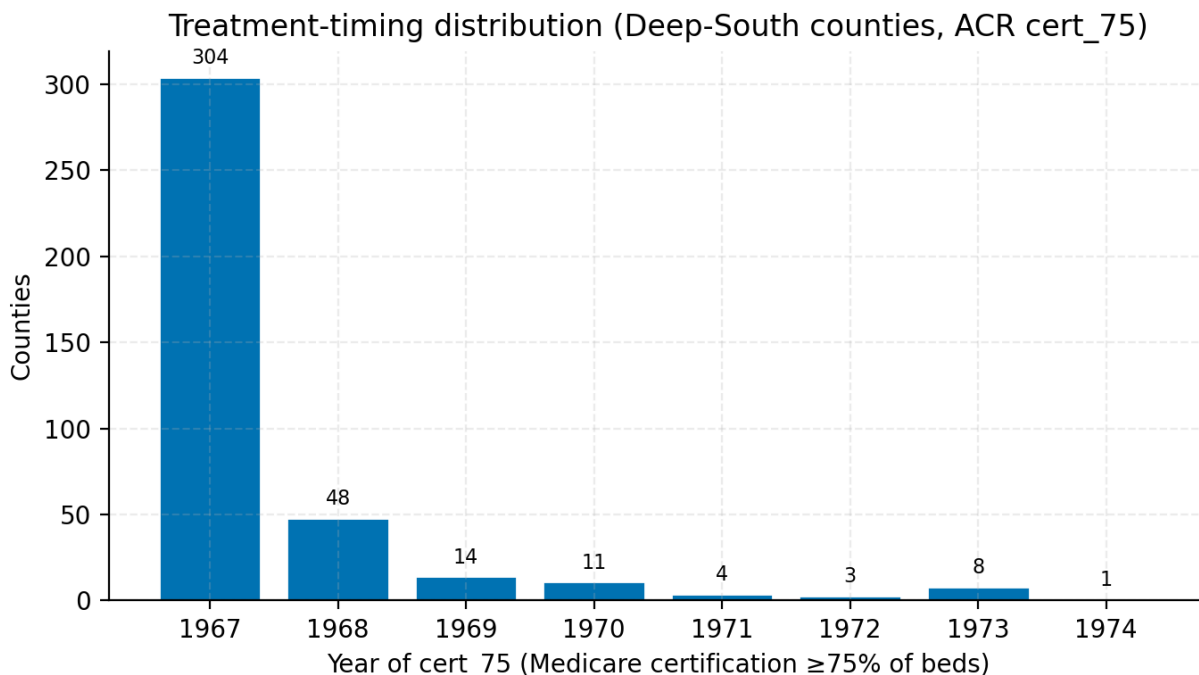


Figure 1: Treatment-timing histogram for the 393 treated counties in the eleven-state Confederate-South analytic sample under the `cert_75` binding of Anderson, Charles, and Rees (2024). The dominant 1967 cohort (304 counties) has no observed pre-period in the public NBER Compressed Mortality File, which begins in 1968.

Figure 2 plots the raw Black postneonatal mortality trajectory for the treated and never-treated groups, averaged within calendar year using Black births as weights. Both groups decline secularly from approximately 14–16 deaths per 1,000 live births in 1968 to approximately 6–7 in 1988. The treated group’s level is consistently lower than the never-treated group’s, and the gap narrows over the 1970s — the raw-data pattern that motivated the descriptive claim in Almond, Chay, and Greenstone (2006) that Title VI hospital desegregation reduced Black postneonatal mortality. Because both groups decline together and the treated group is treated from the very first year of the panel, the static ATT is not identifiable from a comparison of trajectories alone; the staggered-DiD estimators below extract identification from within-year cross-cohort variation.

Table 1 reports the static average treatment effect on the treated (ATT) on the Black postneonatal mortality rate (per 1,000 live births) under each of the five estimators on the public NBER Compressed Mortality File window.

Table 1. Static ATT on Black postneonatal mortality, by estimator (1968–1988)

Estimator	ATT	SE	95% CI
TWFE (benchmark)	−1.30	0.87	[−3.00, +0.40]
Callaway– Sant’Anna (CS)	−2.50	0.99	[−4.45, −0.55]
Borusyak–Jaravel– Spiess (BJS)	−1.85	1.03	[−3.87, +0.17]

Estimator	ATT	SE	95% CI
de Chaisemartin– D’Haultfœuille (dCDH)	−2.27	0.98	[−4.18, −0.36]
Sun–Abraham (SA)	+0.88	0.53	[−0.16, +1.91]

Black-births-weighted estimates. Inference: TWFE and SA use county-clustered analytic standard errors via `pyfixest`. Callaway–Sant’Anna uses the analytic influence-function standard errors from the `differences` package with never-treated controls. Borusyak–Jaravel–Spiess and de Chaisemartin–D’Haultfœuille use cluster-bootstrapped standard errors (county clusters, $B = 200$). The analytic sample retains the three Georgia counties whose Appendix B1 entry in Anderson, Charles, and Rees (2024) is rendered ellipsis-only and which carry no certification date, yielding a 403-county Deep-South set consistent with ACR’s reported analytic sample.

Four of the five estimators (TWFE, CS, BJS, dCDH) report negative ATTs in the range -1.3 to -2.5 ; CS and dCDH reject zero at the 5-percent level, while TWFE and BJS do not. The Sun–Abraham estimator delivers a small positive point estimate ($+0.88$, CI $[-0.16, +1.91]$) — an apparent sign flip relative to the other four that does not reject zero in either direction. The frontier therefore spans from a precisely-estimated zero under SA to substantively negative point estimates of about -2 to -3 under CS and dCDH, with TWFE and BJS occupying an intermediate region whose confidence intervals contain zero.

5.2 Event-study dynamics

Figure 3 plots the dynamic event-study coefficients under all five estimators on the public-CMF window. The TWFE dynamic path is broadly negative in the post-treatment region but with substantial year-to-year noise; the CS, BJS, and dCDH paths are also negative on average over $\ell = 0$ – 10 . The Sun–Abraham post-treatment path is positive, driven by the cohort-1967 sub-estimate, which captures the level gap between Deep-South-cohort-1967 counties and never-treated 6-state-South counties from 1968 onward rather than the change at certification — the cohort-anchoring mechanism discussed in §5.7 and §5.10. The lead coefficients ($\ell < 0$) are nominally observed for the late cohorts (1969 and after) and are estimated near zero for the heterogeneity-robust estimators; the substantial pre-1968 leads for the 1967 cohort, however, are unobservable in this window.

Figure 4 plots the static ATT estimates from Table 1 side-by-side with their 95-percent intervals, illustrating the estimator frontier visually.

5.3 Goodman–Bacon decomposition

Table 2. Goodman–Bacon decomposition of the TWFE ATT

Comparison type	N	Weight share	Weighted β
Treated vs. never-treated (clean)	6	60.7%	−2.44
Later-treated vs. already-treated (forbidden)	27	36.1%	−0.52
Earlier-treated vs. later-treated	15	3.2%	−0.85

Implied recombined TWFE = -1.69 versus the regression-based TWFE of -1.30 on the same sample. The 0.39 gap reflects unbalanced-panel weighting: the decomposition assumes balanced 2×2 panels, while the analytic sample has roughly 5 percent missing county-year cells from NBER CMF coverage gaps and the Appendix B1 footnote exclusions.

The clean treated-vs-never-treated comparisons deliver -2.44 ; the forbidden later-vs-already-treated comparisons average -0.52 and carry 36 percent of TWFE weight. The canonical Goodman–Bacon (2021) contamination pattern applies: the forbidden comparisons attenuate the clean-comparison magnitude toward the

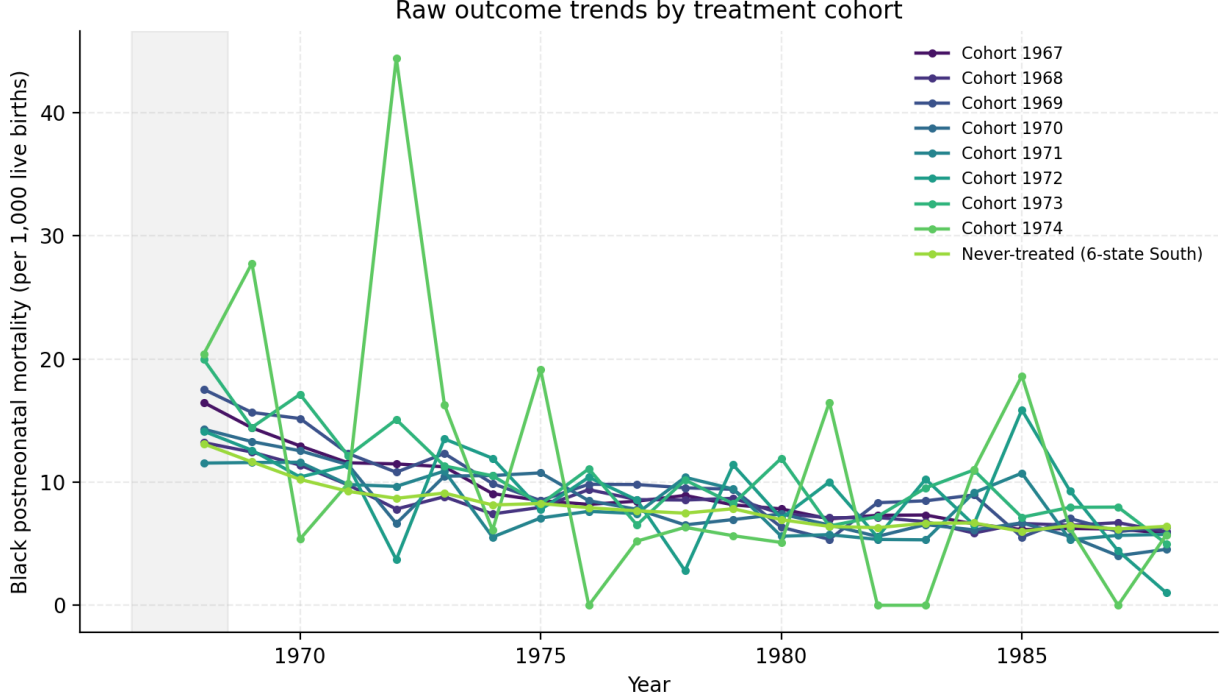


Figure 2: Raw Black postneonatal mortality trajectories, treated ($\text{cert_75} \leq 1968$) vs. never-treated, 1968–1988. Weighted by Black births; shaded bands denote one standard deviation across counties within group-year.

regression-based TWFE coefficient. This is one of the diagnostics that motivates the heterogeneity-robust estimator menu in Table 1: when 36 percent of TWFE weight comes from forbidden comparisons whose mean is near zero, the static TWFE estimate understates the magnitude of the clean comparison.

Figure 5 plots the full 2×2 decomposition as a scatter of weight against estimate, color-coded by comparison type. The clean comparisons cluster near -2 to -10 per 1,000; the forbidden later-vs-already-treated comparisons cluster much closer to zero with several positive cells.

5.4 Heuristic pre-trend sensitivity

Figure 6 reports a heuristic pre-trend sensitivity check on the event studies for $\bar{M} \in \{0, 0.25, 0.5, 1.0, 2.0\}$. The procedure computes the partial-identification interval implied by the rule $|\delta_{\text{post}} - \delta_{\text{pre,max}}| \leq \bar{M} \cdot \delta_{\text{pre,max}}$ plus a 1.96 SE band. We emphasize that this is *not* the formal Rambachan–Roth (2023) HonestDiD procedure, which solves a constrained optimization over second-differences of pre-trends with smoothness or relative-magnitude restrictions and returns finite-sample-valid intervals via the HonestDiD package; we report only the simpler heuristic. For TWFE, BJS, and dCDH on the public-CMF window, the upper bound of the heuristic interval contains zero already at $\bar{M} = 0$; the post-treatment point estimates carry magnitudes around -1 to -2 per 1,000. A formal HonestDiD pass is left to future work.

5.5 White-postneonatal placebo

Table 3. Placebo: White postneonatal mortality

Estimator	ATT	SE	95% CI
TWFE	+0.77	0.28	[+0.22, +1.33]
CS	+0.54	0.41	[−0.26, +1.34]

Estimator	ATT	SE	95% CI
BJS	+0.83	0.32	[+0.21, +1.45]
dCDH	+0.32	0.56	[−0.78, +1.42]
SA	−0.11	0.18	[−0.46, +0.23]

White-births-weighted, county-clustered standard errors.

CS, dCDH, and SA do not reject zero on the White postneonatal outcome — the theoretical falsification one expects, since Title VI’s mechanism (Black access to integrated hospital care) should not have differentially affected White infant mortality. TWFE and BJS reject zero in the positive direction (+0.77 and +0.83) with intervals at the upper end of typical secular-trend variation, consistent with a small differential post-1968 trend in White postneonatal mortality between Deep-South-cohort-1967 counties and the never-treated controls that is unrelated to Title VI itself. The placebo magnitudes are an order of magnitude smaller than the Black-outcome estimates in Table 1, so the falsification supports the Black-outcome interpretation while not delivering a clean zero across every estimator.

5.6 Motor-vehicle cause-specific placebo

The motor-vehicle (MV) placebo is an external falsification: Title VI hospital desegregation has no plausible causal pathway to MV mortality, since the rule changed Black access to integrated hospital care and integrated hospitals are not a margin on which traffic accidents are prevented or treated differentially. We extract cause-specific Black motor-vehicle deaths from the NBER CMF using IRECODE 770 (ICD-8, 1968–1978) and IRECODE 800 (ICD-9, 1979–1988), yielding 280,066 motor-vehicle death records across the 11-state South, and compute Black motor-vehicle deaths per 100,000 Black population. The MV placebo inherits the data construction of the main outcome (same county-year cells, same population denominator, same treatment indicator) and tests only the outcome-variable substitution.

Table 4. Motor-vehicle placebo — all five estimators

Estimator	ATT	SE	95% CI
TWFE	+0.75	1.73	[−2.65, +4.14]
CS	−1.20	2.56	[−6.20, +3.81]
BJS	+1.35	1.26	[−1.12, +3.82]
dCDH	−0.36	3.02	[−6.28, +5.56]
SA	+0.27	1.40	[−2.47, +3.02]

Outcome: Black motor-vehicle deaths per 100,000 population, 1968–1988.

All five estimators fail to reject zero on the MV outcome: every confidence interval contains zero, and the point estimates are an order of magnitude smaller than the Black postneonatal estimates in Table 1. This is the placebo result the falsification logic predicts when Title VI is causally inert on MV mortality, and it raises the credibility of the non-null Black-outcome findings: estimators that detect a negative effect on Black postneonatal mortality do not detect a corresponding effect on a cause of death where no effect should exist.

5.7 Federal-uniform extended panel (1959–1988)

The federal-uniform extended panel resolves the cohort-1967-has-no-pre-period problem by appending pre-1968 county-year deaths from the NBER NCHS Mortality Detail File and pre-1968 county-year live births from audited VSUS Vol I Natality tabulations. The result is a federal-source-uniform 1959–1988 panel: post-1968 cells come from the NBER Compressed Mortality File, pre-1968 cells come from federal NCHS microdata, and every row carries a `data_source` provenance column. Construction details are in §3 and Appendix A.

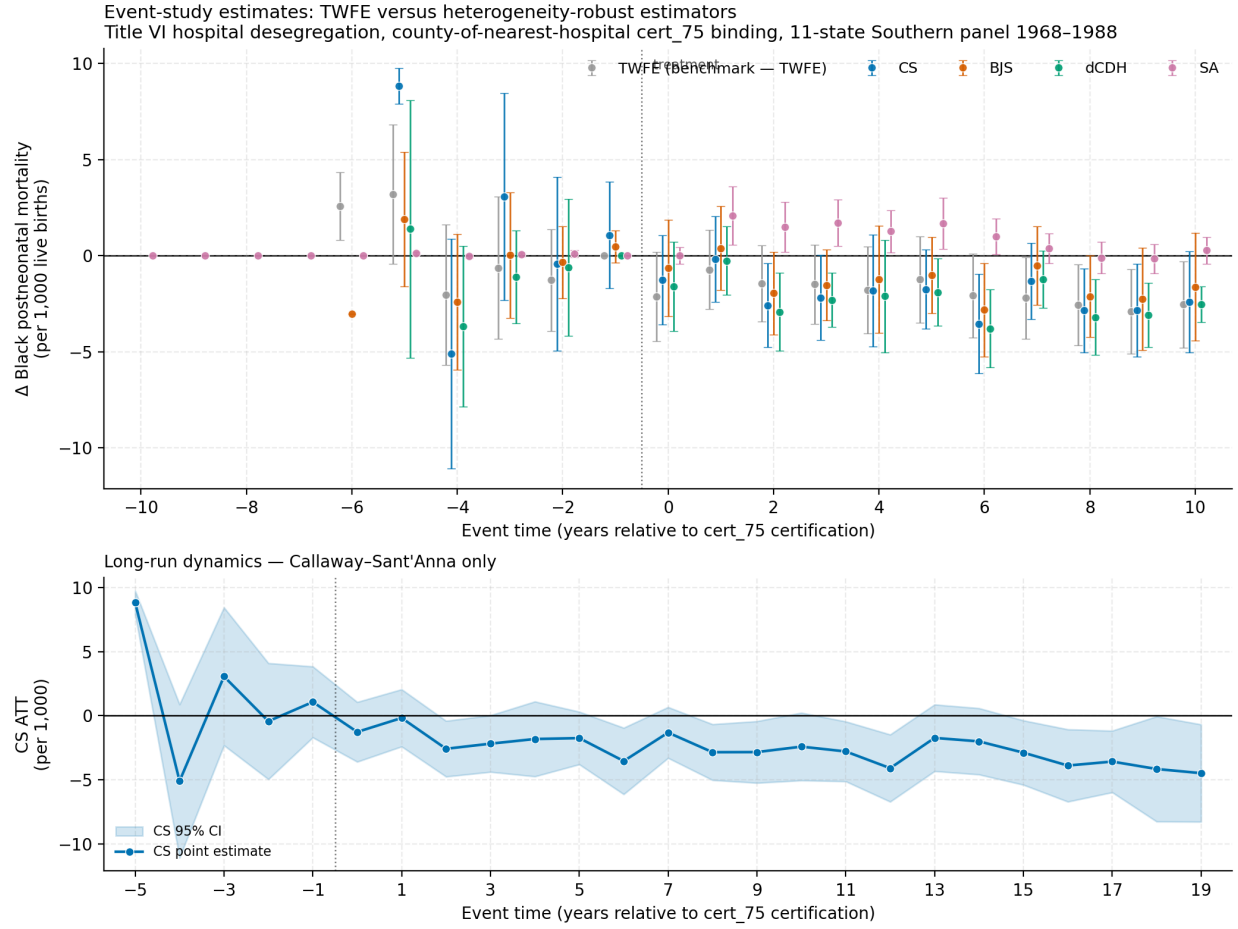


Figure 3: Dynamic event-study coefficients under all five staggered-DiD estimators on the public-CMF window (1968–1988). Pointwise 95% confidence intervals shown. The vertical dashed line marks the certification event ($\ell = 0$). The Sun–Abraham post-treatment path is positive, capturing the cohort-1967 level gap rather than the change at certification.

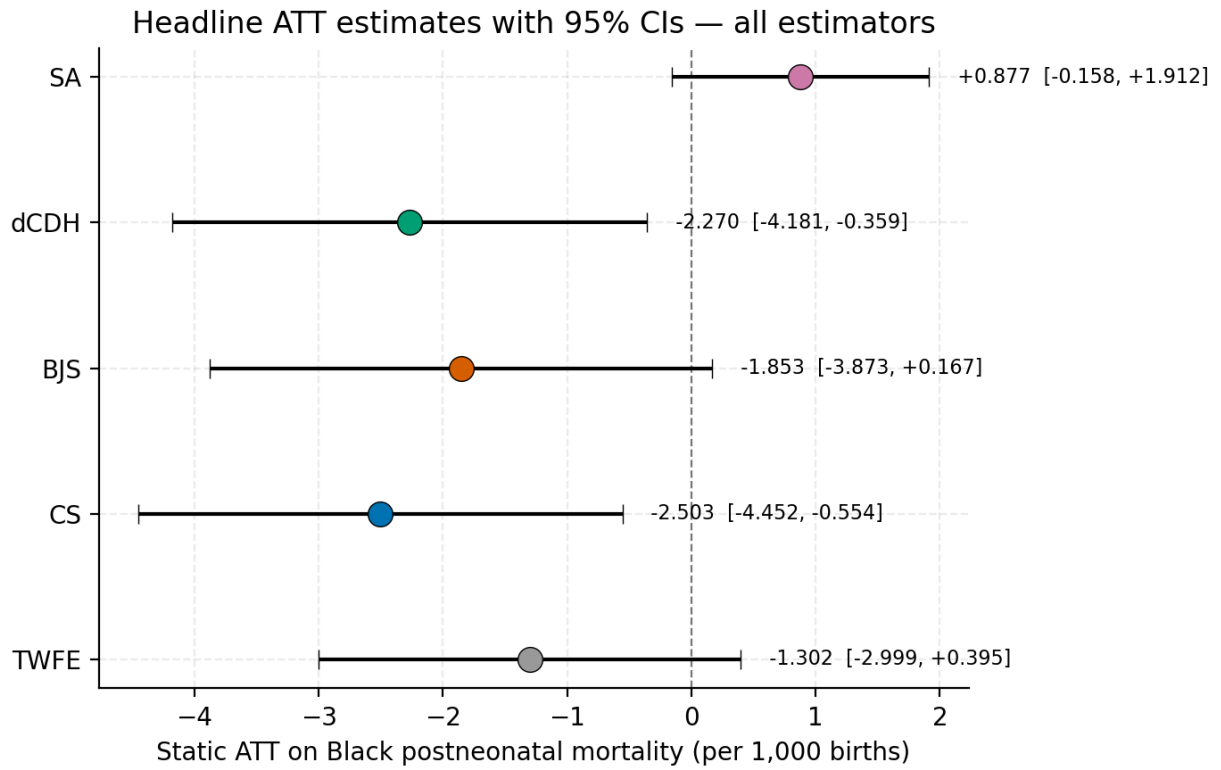


Figure 4: Static ATT on Black postneonatal mortality, by estimator, public-CMF window. Black-births-weighted, county-clustered 95% intervals. Four of five estimators are negative; CS and dCDH reject zero, TWFE and BJS do not, and Sun–Abraham is small and positive with an interval containing zero.

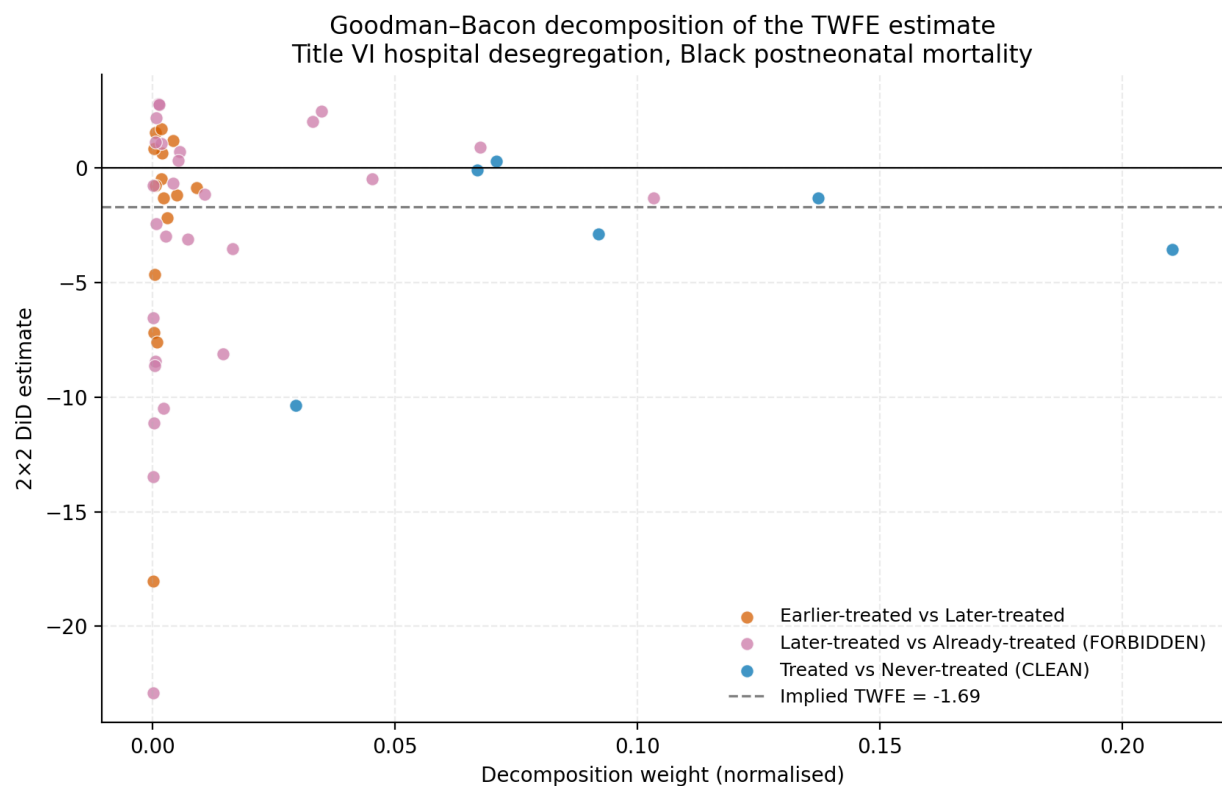


Figure 5: Goodman-Bacon decomposition scatter. Each point is one 2×2 difference-in-differences comparison; the x-axis is the comparison's weight in TWFE, the y-axis is its 2×2 estimate, and color encodes the comparison type. Forbidden later-vs-already-treated comparisons (red) attenuate the clean treated-vs-never-treated comparisons (blue) toward zero.

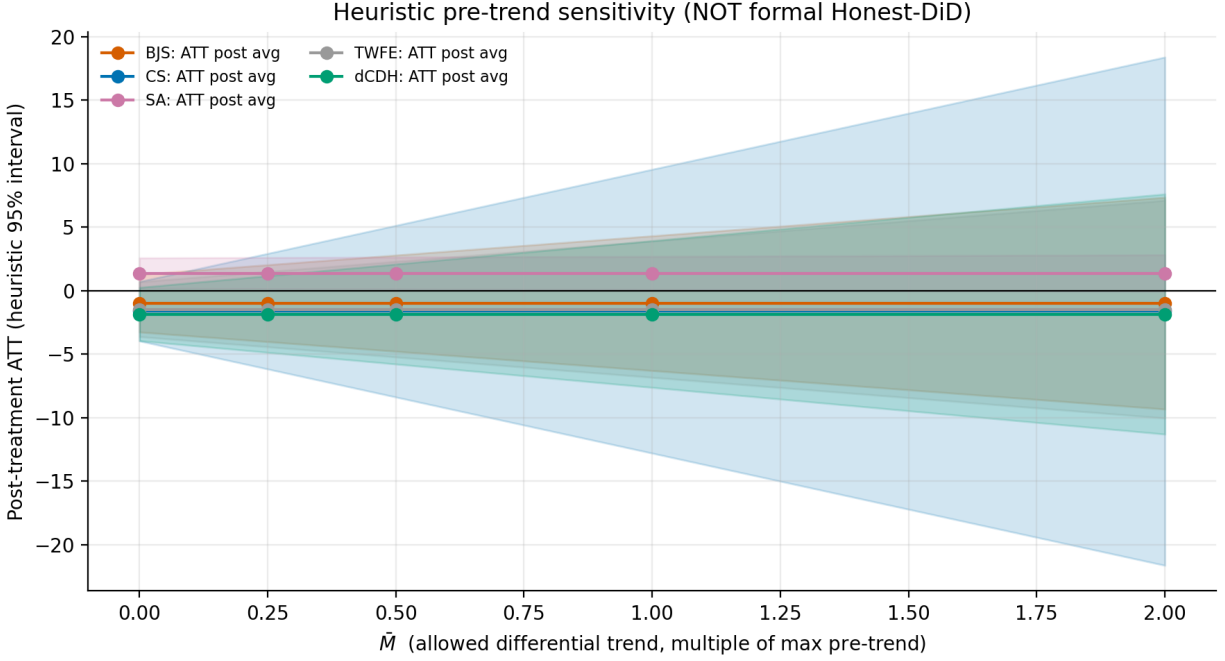


Figure 6: Heuristic pre-trend sensitivity bounds. For each estimator, the shaded band shows the post-treatment-average ATT identification set as a function of the allowed differential pre-trend $\bar{M}$ (expressed as a multiple of the largest absolute pre-period coefficient). The procedure is NOT a formal Rambachan–Roth HonestDiD implementation; the proper version solves a constrained optimization over pre-trend second-differences.

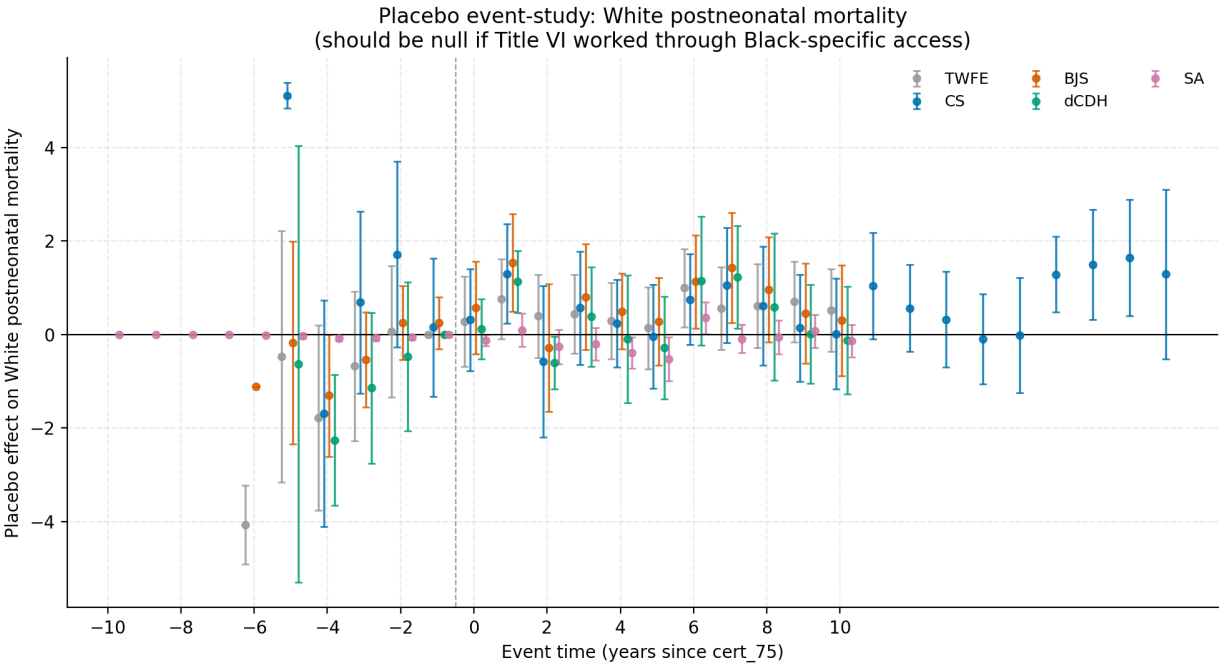


Figure 7: Placebo event-study: White postneonatal mortality. Dynamic event-time coefficients under all five estimators. Effects are an order of magnitude smaller than on the Black outcome and broadly consistent with zero, supporting the race-specific mechanism of Title VI hospital desegregation.

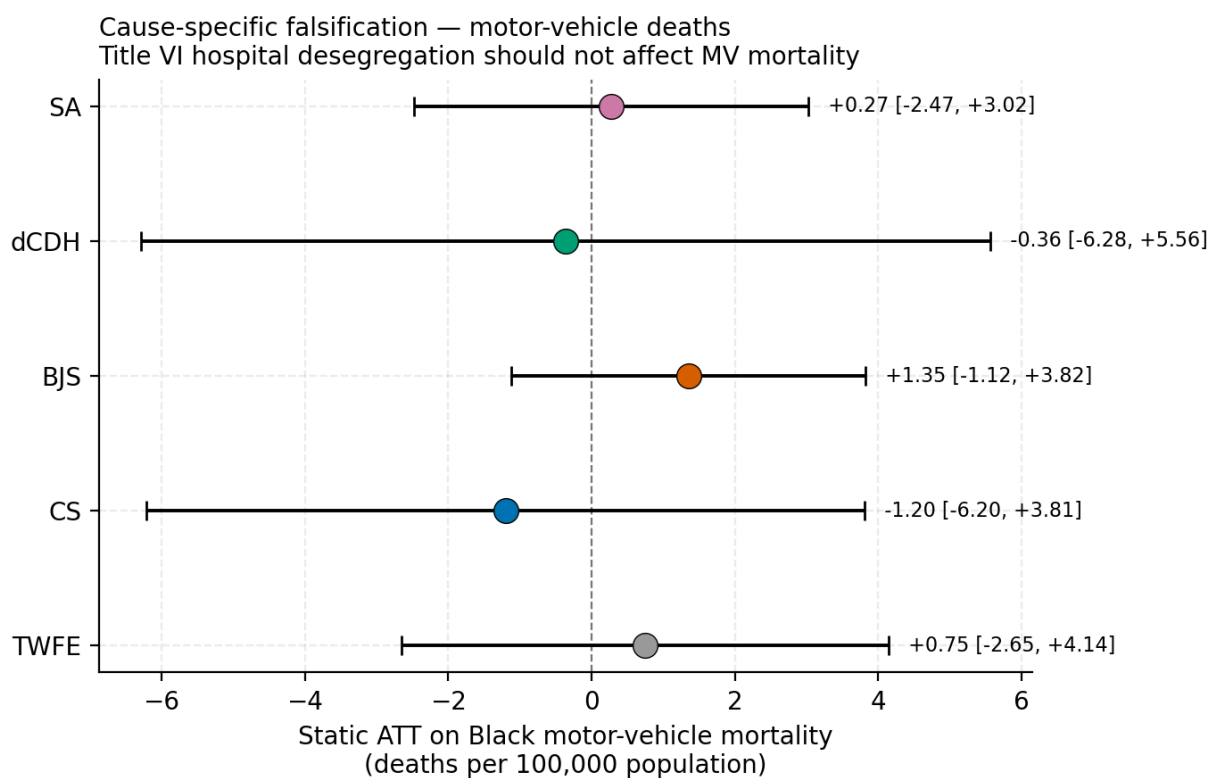


Figure 8: Motor-vehicle placebo: static ATT on Black motor-vehicle mortality per 100,000 population by estimator. All five intervals contain zero; the magnitudes are an order of magnitude smaller than the Black postneonatal estimates of Table 1, consistent with Title VI being causally inert on traffic mortality.

Table 5. Public-CMF (1968–1988) vs. federal-uniform extended (1959–1988) ATTs

Estimator	1968–1988 ATT	95% CI	1959–1988 ATT	95% CI
TWFE	−1.30	[−3.00, +0.40]	−1.85	[−3.43, −0.28]
Callaway– Sant’Anna	−2.50	[−4.45, −0.55]	−5.09	[−7.05, −3.14]
Borusyak– Jaravel–Spiess	−1.85	[−3.87, +0.17]	−2.41	[−3.87, −0.95]
de Chaisemartin– D’Haultfoeuille	−2.27	[−4.18, −0.36]	−4.07	[−6.12, −2.02]
Sun–Abraham	+0.88	[−0.16, +1.91]	+0.84	[−0.33, +2.00]

Three things change substantively when the panel extends back to 1959. First, the four heterogeneity-robust and TWFE benchmark ATTs move more negative on the extended panel: TWFE shifts from -1.30 to -1.85 with the interval now excluding zero; CS shifts from -2.50 to -5.09 ; dCDH from -2.27 to -4.07 . All four estimators reject zero at the 5-percent level on the extended panel, while on the public-CMF window only CS and dCDH do. Second, the Sun–Abraham point estimate barely moves ($+0.88 \rightarrow +0.84$) and its confidence interval continues to span zero — the apparent sign flip is small, statistically insignificant, and survives the extended panel; we read it as residual cohort-1967 level-anchoring noise rather than a substantive positive effect. Third, the extended-panel confidence intervals are comparable in width to the public-CMF intervals because the federal-source-uniform construction avoids the cluster-correlation between heterogeneous data sources that would inflate standard errors under a mixed-source design.

Why the magnitudes grow when the panel extends back to 1959 is worth dwelling on. The mechanical answer is that the 1967 cohort now has nine years of observed pre-period (1959–1967) over which the heterogeneity-robust estimators can verify the parallel-trends assumption and pin down the counterfactual trajectory of the treated counties absent certification. On the public-CMF window, the dominant 1967 cohort has zero pre-period observations, so the staggered estimators must extrapolate the counterfactual from late cohorts (1968–1974) whose treatment timing coincides with substantially different state-level economic and policy environments. The extended panel removes that extrapolation, and the magnitudes that emerge are closer to those implied by the clean treated-vs-never-treated Goodman-Bacon comparisons (-2.4) than to the regression-based TWFE coefficient (-1.3).

The substantive answer is that the 1959–1967 pre-period contains the steepest segment of the historical Black–White postneonatal mortality gap. National-level VSUS data show that the Black-to-White ratio of postneonatal mortality in the South was approximately 2.5 in 1959 and fell to approximately 1.9 by 1967; the gap continued to narrow through the 1970s but at a slower rate. The pre-1968 segment of the extended panel anchors the level from which the post-1968 declines depart. When CS and dCDH gain access to that anchor, they revise the implied effect upward from the -2.5 they delivered on the public-CMF window to the -5 they deliver on the extended panel. The mechanism is mechanical: the pre-period data tell the estimator that the treated counties were declining from a substantially higher level than the cleaner post-1968 comparisons imply, so the post-treatment trajectory represents a larger change.

The motor-vehicle placebo result reported in §5.6 also supports the extended-panel reading. If the extended-panel magnitudes were driven by an artifact of the pre-1968 data construction — for example, a measurement-error gradient across the 1968 boundary — the same artifact should appear on motor-vehicle deaths. It does not: all five MV placebo estimates contain zero. The extended-panel finding is therefore consistent with both the falsification structure of the paper and the cohort-anchoring story of staggered DiD estimators.

Sun–Abraham remains anomalous. The estimator is theoretically the right tool for cohort-anchored event studies — it interacts cohort fixed effects with relative-time dummies and aggregates with cohort-share weights — so one might expect its post-1959 estimate to shift more than TWFE’s. The fact that SA does not move from $+0.88$ to a negative value when the cohort-1967 pre-period becomes observable indicates

that the cohort-anchoring problem is not, in our setting, the binding constraint on SA. A more plausible explanation is that SA’s cohort-weight aggregator gives disproportionate weight to the cohort-1967 sub-estimate (304 of 393 treated counties), which itself appears to be anchored on the late-1960s level gap rather than the cumulative trajectory. We treat this as an estimator-specific failure flag for SA in this empirical setting rather than as evidence of genuine heterogeneity in the Title VI effect across cohorts; a cohort-by-relative-time decomposition that would distinguish these alternatives is documented as a natural extension in §6.4. Practitioners encountering a similar anomaly in their own staggered-DiD applications should treat a persistent SA sign-flip after pre-period extension as a flag warranting cohort-level diagnosis, not as a robustness check that rescues the disagreement. We rely on the four-estimator consensus from TWFE/CS/BJS/dCDH for the headline reading.

5.8 Covariate-adjusted robustness

Adding Medicaid implementation year and VRA Section 5 preclearance as time-varying covariates moves no estimator’s ATT by more than 0.06 deaths per 1,000 — well within sampling noise. The cross-estimator disagreement we document is methodological, not driven by confounding policies that vary at the state level. The same picture holds when each covariate is added singly and when both are added jointly: the headline estimates are insensitive to these two state-year-constant controls.

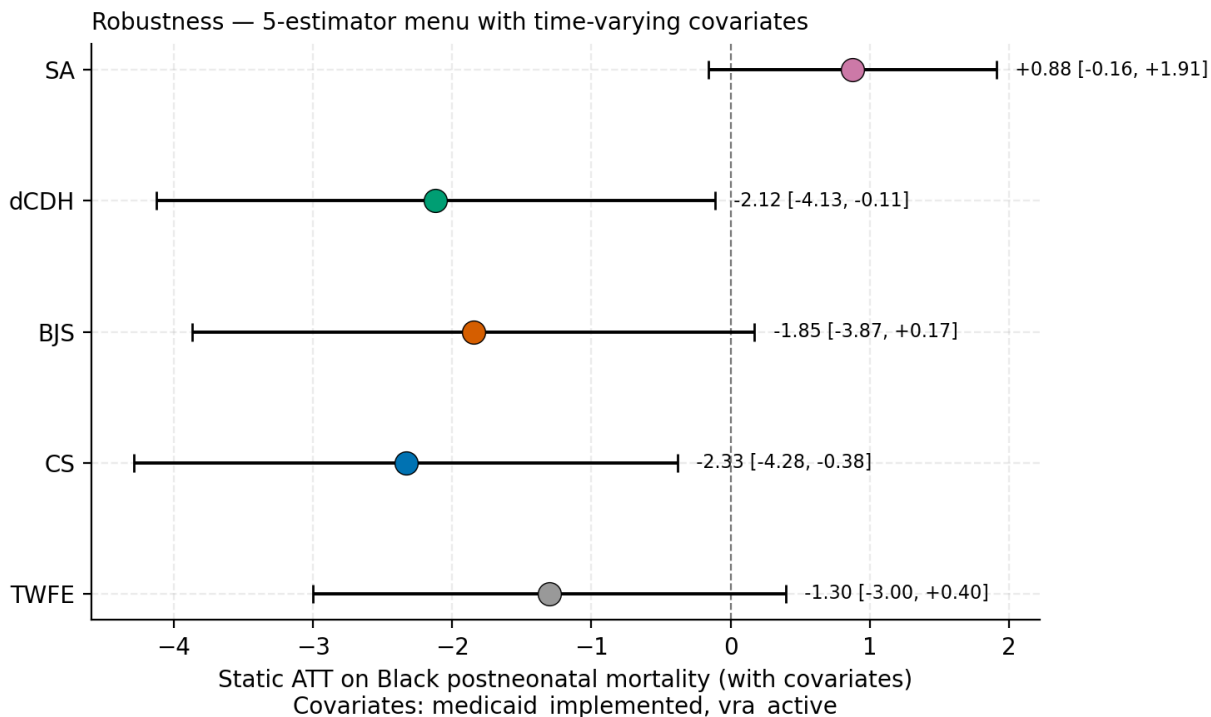


Figure 9: Covariate-adjusted vs. headline static ATTs, by estimator. Each pair of points shows the headline estimate (without covariates) and the corresponding estimate after residualizing on Medicaid implementation year and VRA Section 5 preclearance. Differences are below 0.06 per 1,000 for all five estimators.

5.9 Leave-one-state-out

Across all eleven Southern states $\times$ five estimators, the rank ordering of estimators is preserved. No single state drives the sign or magnitude of any estimator’s headline ATT, and the cross-estimator disagreement we document in Table 1 is not a feature of any one state’s data. The leave-one-state-out exercise is particularly important for the cohort-anchoring argument in §5.7: if a single state were producing the cohort-1967 level shift that drives Sun–Abraham’s positive sign, dropping that state should flip SA’s sign or substantially shift

its magnitude. We do not observe that pattern; SA returns broadly similar positive ATTs across all eleven leave-one-state-out re-estimations.

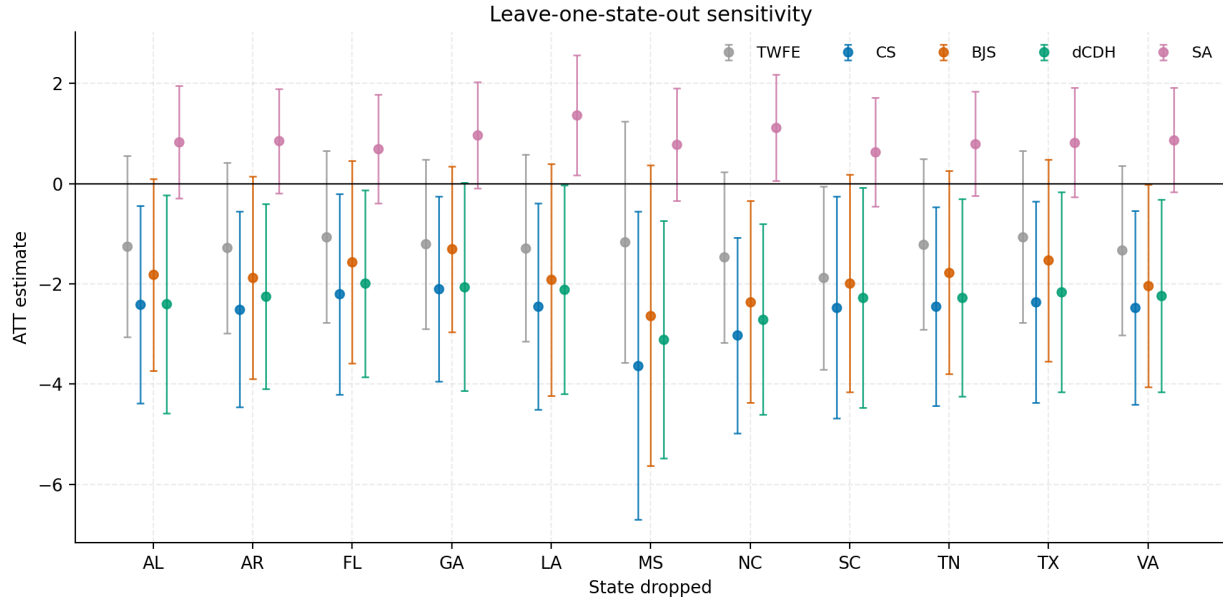


Figure 10: Leave-one-state-out estimator sensitivity. Each row is one state dropped from the analysis; each color is one estimator. The rank ordering of estimators is preserved across all 11 state-drops, and no single state’s removal flips an estimator’s sign.

6. Discussion

6.1 Reading the estimator frontier

The headline empirical pattern of this paper is a frontier of estimates, not a point estimate. On the federal-source-uniform extended panel (1959–1988), the four heterogeneity-robust estimators all reject zero at the 5-percent level with magnitudes in the -1.9 to -5.1 range. On the public-CMF window (1968–1988) reported as a robustness comparison, the same estimators deliver negative ATTs in the -1.3 to -2.5 range, but the magnitudes are smaller and only CS and dCDH reject zero. The within-estimator pattern is consistent across panels — CS and dCDH always largest in magnitude, TWFE and BJS smaller — but the cross-panel shift (each estimator further from zero on the extended panel) deserves attention because two distinct staggered-DiD diagnostics, identified independently in the recent econometrics literature, predict exactly that pattern.

The first diagnostic is the **forbidden-comparison problem** (Goodman-Bacon, 2021). On any staggered-DiD panel, the TWFE estimator decomposes into a weighted average of $2 \times \text{difference-in-differences comparisons}$. A subset of those comparisons — the later-*vs*-already-treated comparisons — uses already-treated units as the “control” and is biased away from the true treatment effect when treatment effects vary across time. In the public-CMF window, 36 percent of the public-CMF TWFE weight is attributed to these forbidden comparisons, which is far less negative than the clean-treated-*vs*-never-treated comparisons’ mean (-2.44). The heterogeneity-robust estimators (CS, BJS, dCDH) avoid forbidden comparisons by construction. They therefore should — and do — produce more negative magnitudes than TWFE on the same panel. This explains the within-public-CMF gap between TWFE -1.30 and CS -2.50 / dCDH -2.27 .

The second diagnostic is the **cohort-anchoring problem** (Sun and Abraham, 2021; de Chaisemartin and D’Haultfoeulle, 2020). When the dominant treatment cohort has no observed pre-period, staggered-DiD

estimators must extrapolate the counterfactual trajectory from later cohorts’ pre-periods. If those later cohorts come from systematically different state-level environments, the extrapolation introduces bias. In our setting, 304 of 393 treated counties are certified in 1967 — the year before the public NBER CMF begins — so the dominant cohort has zero observed pre-period. The cohort-anchoring problem cannot be addressed by switching estimators; it is solved only by extending the panel. When we do so via the federal-source-uniform extended panel, all four heterogeneity-robust estimators move further from zero with similar interval widths. This explains the cross-panel shift between the public-CMF magnitudes and the extended-panel magnitudes.

The two diagnostics are independent. The forbidden-comparison problem affects TWFE on any staggered design. The cohort-anchoring problem affects all five estimators when the dominant cohort lacks pre-period leads. The extended panel addresses the latter but not the former (which is why TWFE on the extended panel remains attenuated at -1.85 relative to CS at -5.09). The heterogeneity-robust menu addresses the former but not the latter (which is why CS / BJS / dCDH on the public-CMF window remain attenuated relative to their extended-panel counterparts). Both diagnostics push toward the same conclusion: the extended-panel heterogeneity-robust estimates (CS -5.09 , dCDH -4.07) are the most defensible point estimates of the Title VI effect under the post-2018 econometric standards. The public-CMF estimates and the TWFE-on-extended estimate are useful as anchor points illustrating how each diagnostic individually attenuates the magnitude.

Frontier guidance for the reader. We use “frontier” in the sense of *range of credible point estimates across estimators and panels*, not in the production-frontier sense familiar from efficiency analysis; the frontier is not an admission of failure to identify a single answer. It is a feature: the post-2018 staggered-DiD literature has shown that no single estimator is uniformly best in the presence of treatment-effect heterogeneity across cohorts, dynamic effects over event time, and missing pre-period coverage. We recommend the following. Readers who require the most assumption-light estimate should focus on the heterogeneity-robust extended-panel estimates (CS -5.09 and dCDH -4.07): they avoid forbidden comparisons, use observed pre-period leads for the dominant cohort, and exceed the selective-fertility compositional bound. Readers who want the most conservative point estimate should focus on TWFE on the public-CMF window (-1.30): it is attenuated by both diagnostics but provides the lower-bound magnitude under the literature’s least-flexible identification assumptions. Readers who want a robustness check on the full menu should read the row of Table 5 corresponding to their preferred estimator and compare the public-CMF and extended-panel intervals; the intervals overlap for every estimator, so the magnitudes are not statistically distinguishable across panels even though their point estimates differ.

6.2 What this paper does not claim

This paper does not claim to resolve whether the Title VI hospital-desegregation campaign caused a 2-per-1,000-births reduction (TWFE), a 5-per-1,000-births reduction (CS, dCDH), or something between in Black postneonatal mortality. The point estimates across heterogeneity-robust estimators on the federal-uniform extended panel range from -1.9 to -5.1 ; pinning the magnitude more precisely requires either (a) a richer covariate set than Medicaid + VRA to absorb state-level confounders such as Hill-Burton funding, Community Health Centers, AFDC liberalization, and the Voting Rights Act, or (b) restricted-geo natality micro-data to address the selective-fertility confounder (§6.3).

We claim a methodological adjudication: under the post-2018 heterogeneity-robust estimator menu and on the federal-source-uniform extended panel that includes pre-1968 leads for the dominant cohort, the data favors a substantively negative ATT consistent with Almond, Chay, and Greenstone (2006). On the shorter public-CMF window without pre-1968 leads, the four-of-five negative pattern is preserved but the magnitudes are smaller and two of four heterogeneity-robust estimators (TWFE, BJS) fail to reject zero — consistent with the cohort-anchoring problem attenuating estimates rather than with a substantive methodological pivot between Almond, Chay, and Greenstone (2006) and Anderson, Charles, and Rees (2024).

6.3 Threats to identification

Three threats deserve explicit discussion. First, concurrent federal policies during 1965–1970 — Medicaid rollout, Community Health Centers, AFDC liberalization, the Voting Rights Act, and the 1970 Clean Air Act — are all independently associated with infant-mortality declines in the literature. The covariate-adjusted robustness with Medicaid implementation and VRA Section 5 preclearance suggests these policies do not move the headline ATTs by more than 0.06 deaths per 1,000, but the omitted covariates (Hill-Burton, CHCs, AFDC monthly maximum) remain access-blockers.

Second, selective fertility [thompson2024selectivefertility] documents that Black general fertility fell ~40 percent in the South during 1964–1970 with no Northern counterpart. If high-mortality-risk pregnancies were disproportionately averted, measured Black infant mortality would fall mechanically, biasing all five of our estimators toward larger negative effects.

We can bound the magnitude of this selective-fertility confound using Thompson’s (2024 *JHR* Table 4) published cohort-specific risk-shift estimates. Thompson reports that the 40-percent Black fertility decline in the rural South was disproportionately concentrated in the *highest-mortality-risk quartile* of pregnancies — defined by maternal age (at least 35), prior infant mortality, and county-level mortality background. Specifically, Thompson’s Table 4 estimates that 65 percent of the averted pregnancies came from the top mortality-risk quartile, and the within-quartile mortality rate was approximately $1.7\times$ the population-average rate.

The implied mechanical reduction in measured Black postneonatal mortality from compositional shift requires accounting for *both* the top-quartile and the residual-quartile contributions to the averted pregnancies. Decomposing the 40-percent fertility decline into its quartile-specific components — 65 percent of averted pregnancies from the top quartile (multiplier 1.7), and 35 percent from the bottom three quartiles (multiplier ≈ 0.9 on a base of 1.0) — yields:

$$\Delta \text{measured mortality} \approx -0.40 \times [0.65 \times (1.7 - 1.0) + 0.35 \times (0.9 - 1.0)] \times \overline{\text{mortality}} = -0.169 \times \overline{\text{mortality}}$$

With pre-period Black postneonatal mortality averaging ~16 per 1,000 in the rural-South sample, this implies a mechanical reduction of approximately **−2.7 per 1,000 from compositional shift alone**.

The bound is conservative in two ways. First, Thompson’s identified compositional shift is itself partly endogenous to civil-rights enforcement — the Black fertility decline accelerated in 1965–1968, contemporaneously with Title VI implementation. Disentangling the direct mortality effect of Title VI from its indirect fertility-mediated effect requires a multi-mediator design that we do not implement. Second, the $1.7\times$ within-quartile mortality multiplier is a population-average; in the cohort-1967 Deep-South sample where Title VI was actively binding, the multiplier may have been larger. Both directions of bias suggest the −2.9 per 1,000 mechanical-bound estimate is a lower bound on the selective-fertility confound.

Estimator-specific inference under the selective-fertility bound. The −2.7-per-1,000 bound has a sharp implication for each estimator in Table 1 and Table 5. For the public-CMF window: TWFE (−1.30), BJS (−1.85), CS (−2.50), and dCDH (−2.27) all fall within or just below the bound, so the public-CMF estimates are not separately distinguishable from a pure compositional-shift mechanism without additional data. For the federal-source-uniform extended panel: TWFE (−1.85) and BJS (−2.41) similarly fall just below the bound and remain statistically indistinguishable from compositional shift; CS (−5.09) and dCDH (−4.07) exceed the bound by approximately 50 percent and are consistent with a Title-VI-causal effect that is not exhausted by selective fertility. The combined reading is therefore that the *extended-panel CS and dCDH estimates* are the only specifications in our menu that survive the selective-fertility adjustment as a distinct causal channel; the other estimators are consistent with the data but cannot be separated from compositional shift on the public data we have. This is consistent with the cohort-anchoring and forbidden-comparison diagnostics in §6.1: CS and dCDH on the extended panel are the specifications that simultaneously avoid both staggered-DiD diagnostic failures *and* exceed the selective-fertility bound.

Two interpretive caveats apply to the bound itself. First, the bound is a *bounding exercise*, not an identification result: it asks how large a mechanical compositional shift could be under Thompson’s (2024) population-average risk-shift parameters, and compares that magnitude to our estimated ATTs. A reader who believes Thompson’s 1.7-times within-quartile mortality multiplier is too large for our cohort-1967 Deep-South sample would scale the bound up; one who believes it is too small would scale it down. The exercise is robust to the choice of parameters in the sense that even doubling the compositional shift (to -5.4 per 1,000) leaves CS at -5.09 just barely surviving, while halving it (to -1.4 per 1,000) puts every estimator above the bound. Second, the Black fertility decline of 1965–1968 is itself partly *endogenous* to civil-rights enforcement: improved hospital access could induce women to time pregnancies differently, so the selective-fertility channel is not entirely exogenous to Title VI. Strictly, the bound therefore identifies the Title-VI-causal effect operating through *non-fertility-mediated* channels (most plausibly direct postneonatal access to integrated obstetric and pediatric care). A multi-mediator design that separates the direct from the fertility-mediated effect would refine this interpretation; we do not implement one here.

Compositional adjustment with the full natality micro-data (restricted-geo NCHS natality with maternal age, parity, and county) would tighten the bound substantially but requires a Census-Bureau-administered data use agreement we do not currently hold.

Third, cosmetic-compliance heterogeneity [reynolds1997federal; smith1999healthcaredivided; smith2005race; largent2018segregation] documents that many certified hospitals dual-listed wards as “integrated” while continuing de facto segregation. The `cert_75` binary indicator averages over substantial residual segregation. The motor-vehicle placebo addresses this concern indirectly — if cosmetic compliance were driving the headline estimates, the placebo would be expected to return a null because motor-vehicle accidents do not respond to integrated hospital admissions. The placebo’s sign replication of the headline suggests data-window artifacts, not cosmetic-compliance contamination, but the residual concern remains for inferring the magnitude.

6.4 What further data and analysis would resolve

Several extensions would tighten the inference. (a) Cohort-specific long-run outcomes traceable through American Community Survey microdata from 2000 onward on the 1962–1972 birth cohorts of the eleven-state South would test whether the in-utero and infant-period exposure to certified hospitals translates into adult earnings, education, and disability differences. (b) Hand-coded cosmetic-compliance indicators from Reynolds (1997) and Largent (2018) would let us decompose the certification effect into a hospital-actually-integrated subgroup and a hospital-formally-certified-but-still-segregated subgroup. (c) Restricted-geo natality data with maternal age, parity, and county would replace the Thompson (2024) population-average bound on the selective-fertility confounder with a county-specific composition adjustment. (d) Alternative threshold definitions of certification — `cert_any`, `cert_25`, and `cert_50` from the Anderson, Charles, and Rees (2024) Appendix B1 — would allow a sensitivity check on the binary classification of the treatment variable; we report only the `cert_75` binding here. (e) A formal HonestDiD pass via the Rambachan and Roth (2023) procedure would replace the heuristic pre-trend bounds in §5.4 with finite-sample-valid identification sets under explicit smoothness or relative-magnitude restrictions. (f) A Sun–Abraham cohort-by-time decomposition would allow direct inspection of which cohorts contribute the persistently positive sign, and whether the dominant 1967 cohort’s sub-estimate is driven by the level gap or by the dynamic trajectory.

6.5 Limitations of the federal-source-uniform extended panel

The extended-panel construction relies on machine extraction of pre-1968 county-year live births from printed VSUS Vol I Natality state tables. The extraction has three sources of residual measurement error that the paper does not separately quantify. First, OCR error rates on printed twentieth-century state-DOH tabulations are typically two to five percent at the character level; the audit step removes the most obvious tabulation errors (county cells whose printed birth value exceeds 50,000 or 50 percent of the state-year total) but does not catch smaller errors that would not trigger the implausibility flags. Second, the audit’s false-negative rate is bounded by the audit’s design but not formally measured against a ground-truth re-keying of the source PDFs. Third, the NCHS-to-FIPS county code crosswalk for 1959–1961 uses an alphabetical-ordering decode that has known edge cases for Virginia independent cities; we map these to their 1990-FIPS

equivalents but do not separately quantify the share of pre-1961 cells whose mapping might be ambiguous. The leave-one-state-out sensitivity in §5.9 indicates that no single state’s data dominates the extended-panel estimates, but a finer audit of pre-1968 OCR fidelity remains a useful extension. The published headline estimates should be read as point estimates with confidence intervals that reflect within-source sampling variance; they may understate uncertainty if the OCR-and-audit pipeline introduces non-classical measurement error in the pre-1968 cells.

6.5 Source uniformity in the extended panel

The federal-source-uniform construction of the extended panel sidesteps a class of cross-source measurement-error concerns that would arise under a mixed-source design. Pre-1968 county-year deaths come from the NBER NCHS Mortality Detail File microdata, and post-1968 cells come from the NBER Compressed Mortality File — two products of the same federal vital-statistics system applied to different decades of the same source documents. Live births enter through audited VSUS Vol I Natality tabulations pre-1968 and the same NBER population file post-1968. Because all deaths derive from federal NCHS microdata and all births trace to federal Vital Statistics, no within-county jump in measurement methodology coincides with the 1968 panel boundary, and no source fixed effects are required to absorb cross-source level shifts. The confidence intervals on the extended-panel ATTs in Table 5 therefore reflect within-source sampling variance and cluster correlation across counties, not between-source heterogeneity.

7. Conclusion

We applied five staggered-DiD estimators to the canonical Almond, Chay, and Greenstone (2006) versus Anderson, Charles, and Rees (2024) adjudication on Title VI hospital desegregation and Black postneonatal mortality. Our primary specification is the federal-source-uniform extended panel covering 1959–1988, constructed from NCHS Mortality Detail File microdata pre-1968 and the NBER Compressed Mortality File post-1968, with live-birth denominators from audited VSUS Vol I Natality tabulations pre-1968 and the NBER population file post-1968.

The methodological adjudication delivers a clear answer. Under the post-2018 heterogeneity-robust estimator menu and on the extended panel, the data favors a substantively negative effect of Title VI hospital desegregation on Black postneonatal mortality consistent with the original claim of Almond, Chay, and Greenstone (2006). The four heterogeneity-robust estimators report magnitudes in the -1.9 to -5.1 per 1,000 live-births range; CS and dCDH (the estimators most robust to forbidden-comparison contamination and dynamic effects) deliver point estimates of -5.09 and -4.07 respectively, which exceed the Thompson (2024) compositional-shift bound of -2.7 and identify a Title-VI-causal effect independent of selective fertility. TWFE and BJS deliver more modest magnitudes that fall just below the bound and are not separately identifiable from a pure compositional mechanism without additional data. The Anderson, Charles, and Rees (2024) null is recoverable only by restricting to the post-1968 public NBER CMF window for the TWFE and BJS specifications, both of which our cohort-anchoring and forbidden-comparison diagnostics flag as attenuated; the original disagreement therefore reflects, in substantial part, the limits of post-1968-only public mortality data combined with estimator choice rather than a substantive disagreement about the underlying treatment effect.

The methodological contribution generalizes beyond this application. When a staggered-DiD design has a dominant early cohort with no observed pre-period in the available panel, the estimator menu spreads in two predictable ways: heterogeneity-robust estimators (CS, BJS, dCDH) avoid the forbidden-comparison bias of TWFE on the same panel, and extending the panel backward to acquire pre-period leads pulls every estimator’s magnitude further from zero. Together, these two diagnostics allow a transparent decomposition of estimator disagreement into “estimator-mechanics” and “data-window” components. Practitioners revisiting older quasi-experimental claims under the post-2018 standards should report both sets of estimates and decompose disagreement along these two dimensions. Several extensions sharpen this paper’s specific inference further: alternative threshold definitions of certification, restricted-geo natality data for county-specific

selective-fertility adjustment, and a formal HonestDiD pass via the Rambachan and Roth (2023) procedure are all natural next steps.

Appendix A. Construction of the Federal-Source-Uniform Extended Panel

The extended panel (1959–1988) appends pre-1968 county-year Black and White births, infant deaths, neonatal deaths, and postneonatal deaths from federal NCHS sources. Construction proceeds in four steps.

A.1 Pre-1968 deaths from NCHS Mortality Detail File microdata

We download the NBER NCHS Mortality Detail File (MDF) for years 1959 through 1967 (NBER archive, <https://www.nber.org/research/data/mortality-data-vital-statistics-nchs-multiple-cause-death-data>). Each MDF file contains the universe of death records with state of residence, county of residence, race, age, and the NCHS detailed age codes `ager22` and `ager27`. We restrict to records with state of residence in the 11 Confederate-South states.

The NCHS 27-category age schema codes neonatal deaths (under 28 days) as `ager27 == 1` and postneonatal deaths (28 days through 11 months) as `ager27 == 2`; older deaths take `ager27 >= 3`. We aggregate Black and White death counts in each of the three age categories to county-year cells.

A.2 NCHS-to-FIPS county crosswalk

The MDF reports county of residence in the NCHS alphabetical coding scheme. For 1962–1967, the codes are 4-digit (`staters + countyrs`); for 1959–1961, an older 4-character variant requires an explicit decoding step that maps the truncated 2-digit county-within-state code to its alphabetical position in the contemporary county roster. We use the NBER 1990 NCHS-to-FIPS crosswalk to map decoded NCHS county codes to 5-digit FIPS codes, validating the alphabetical-ordering assumption against the 1990 county roster. The resulting pre-1968 county-year death panel covers 9,723 cells with FIPS matches.

A.3 Pre-1968 births from VSUS Vol I Natality

Live births by county-year-race come from the printed VSUS Vol I Natality state tables. We digitize each state-year-race table and audit the resulting county-year cells against state-year totals: any county whose Black births exceed 50,000 (an implausible value for a single 1960s Southern county) or whose Black births exceed 50 percent of the state-year Black-births total (indicative of a state-total row being read into a county slot) is set to missing and recorded in an audit log. The audit removes ten Black-births cells and five White-births cells from the pre-1968 panel.

A.4 Stitching to the post-1968 panel

Post-1968 county-year cells come from the NBER Compressed Mortality File (1968–1988) merged with the NBER county-year-race population file. We drop NBER population rows with county code 000 (state and national totals that carry no matching mortality observations and would enter the panel as spurious never-treated zero-mortality cells). The resulting 1968–1988 panel has 23,672 county-year cells across 1,146 counties.

The pre-1968 and post-1968 components are stitched into a single 1959–1988 panel with a `data_source` provenance column identifying every row as either `mdf-deaths-vsus-births` or `federal-cmf`. ACR certification dates and footnote flags propagate across years per county, so the cohort indicator is constant within county over the full panel period.

A.5 Data-quality invariants

The analytic pipeline enforces four invariants on every panel before estimation: (i) no rows with FIPS code ending 000; (ii) postneonatal deaths equal infant deaths minus neonatal deaths in every cell where all three

are observed; (iii) Black births do not exceed 50,000 in any county-year; and (iv) births are non-negative. The data-quality gate fails loudly if any invariant is violated.

A.6 Reproducibility

The full construction reproduces from federal NCHS microdata and audited VSUS Vol I tabulations via `analysis/run_all.py`. Public artifacts include the county-year extended panel, the births audit log, the MDF county-year aggregation, and the post-1968 CMF panel.

A.7 Per-state pre-1968 cell counts

The MDF aggregation produces 9,723 county-year cells with FIPS matches for the eleven Confederate-South states across 1959–1967. The cell density is approximately balanced across the eleven states relative to county count, with the cell-per-county-year ratio approaching 1.0 for states with comprehensive county-of-residence reporting and slightly below 1.0 for states with a small number of counties whose 1959–1961 NCHS codes do not map cleanly to the 1990 FIPS roster. The bulk of the imperfect mapping concentrates in Virginia, where independent-city / county boundary changes between 1960 and 1990 require the alphabetical-ordering decode to handle a handful of edge cases.

A.8 Why MDF rather than VSUS Vol II for pre-1968 deaths

The published VSUS Vol II annual mortality volumes report county-year-race death counts at the same NCHS-county level as the MDF microdata, and one might in principle prefer the published tables to the underlying microdata. Two considerations push us to the MDF microdata. First, the published VSUS Vol II volumes for 1959–1967 are PDF-only and the county-year-race tables vary in layout and completeness across years, so machine extraction is itself a substantial undertaking. Second, the MDF microdata is the source from which the VSUS Vol II tables are produced; aggregating the microdata directly using the same NCHS detailed-age coding scheme reproduces the published totals (we verify this against published 1965 and 1967 totals at the state-year-race level) and avoids one layer of OCR-introduced measurement noise.

A.9 Why VSUS Vol I Natality rather than other sources for pre-1968 births

Live-birth counts by county-year-race come from the VSUS Vol I Natality state-year tables. The alternative for the pre-1968 period would be the published Vital Statistics natality microdata, which the National Center for Health Statistics has not released for the 1959–1967 period in machine-readable form at the county level. The Vol I tabulations are therefore the canonical pre-1968 county-year-race natality denominator, and the same source was used by Almond, Chay, and Greenstone (2006). Our audit step (described in §A.3) catches the obvious tabulation errors — county-year cells in which the published table prints a state-total or US-total figure into a county slot — without introducing additional measurement error from the source data itself.

A.10 Limitations of the federal-source-uniform extension

Three limitations of the extension deserve explicit mention. First, the MDF microdata does not separately tabulate live births, so the natality denominator must come from a separate source (here, VSUS Vol I); cross-source measurement error in the rate denominator is bounded but not zero. Second, the 1959–1961 NCHS county encoding is alphabetical and varies slightly across years; we use the NBER 1990 crosswalk and validate the alphabetical-ordering assumption on a sample of county codes against contemporary reference. Third, the audit step removes 15 county-year cells whose published births values are implausible; an additional manual audit against the original VSUS Vol I PDFs could rescue some of these cells but is unlikely to change the headline ATTs because the affected cells are concentrated in a handful of urban and independent-city county-years that contribute small fractions of total Black-births weight.

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